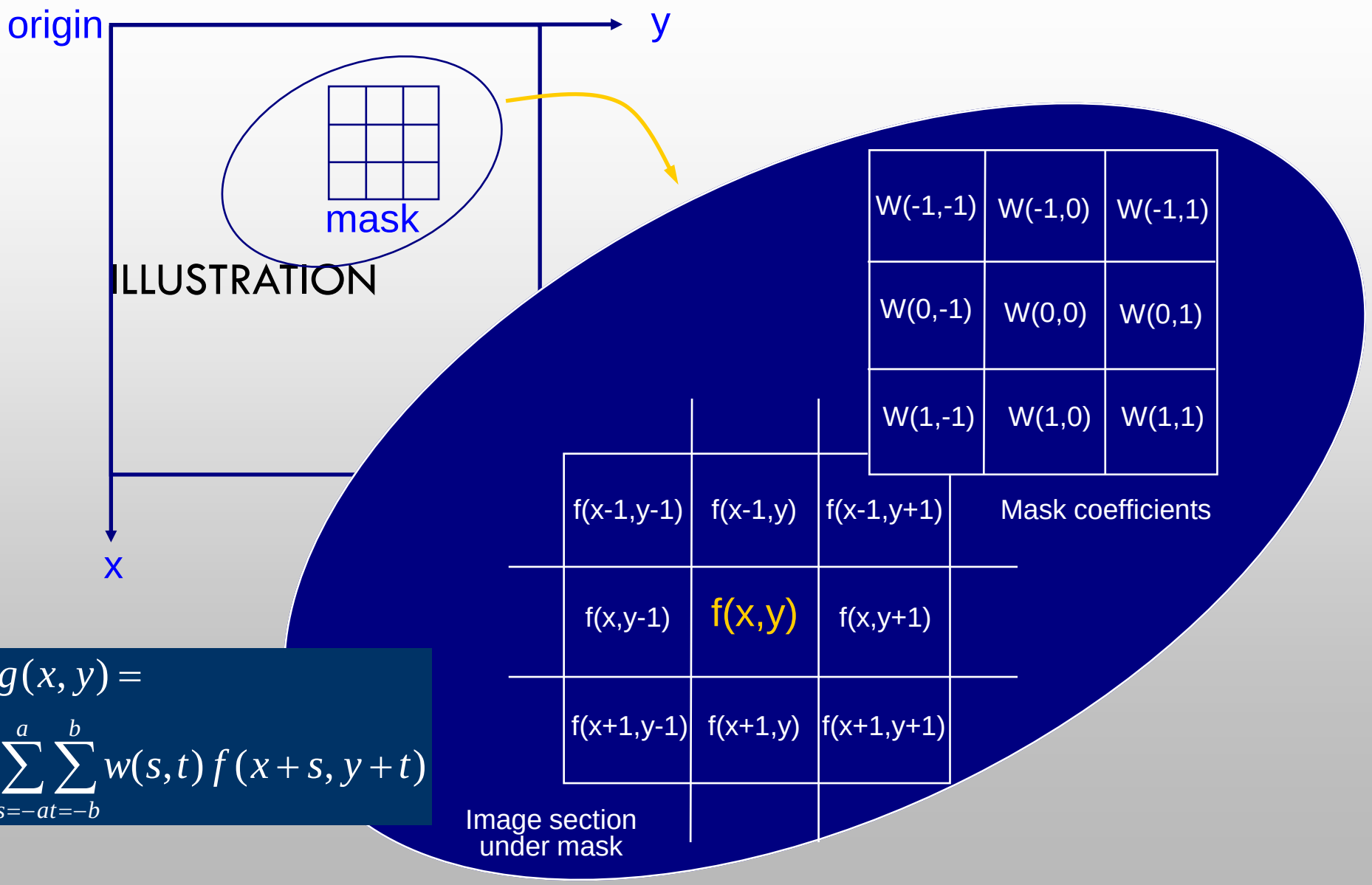
SPATIAL FILTERING

- DEFINITION: A PROCESS THAT MOVES A SUBIMAGE FROM POINT TO POINT IN AN IMAGE, WITH THE RESPONSE AT EACH IMAGE POINT PREDEFINED.
- THE SUBIMAGE: FILTER, MASK, KERNEL, TEMPLATE OR WINDOW.
- THE VALUES IN A FILTER SUBIMAGE ARE CALLED *COEFFICIENTS*, NOT PIXELS.
- THE PROCESS IS ALSO NAMED CONVOLUTION.
- CONVOLUTION OF 2D FUNCTION f AND IS DENOTED w OR g

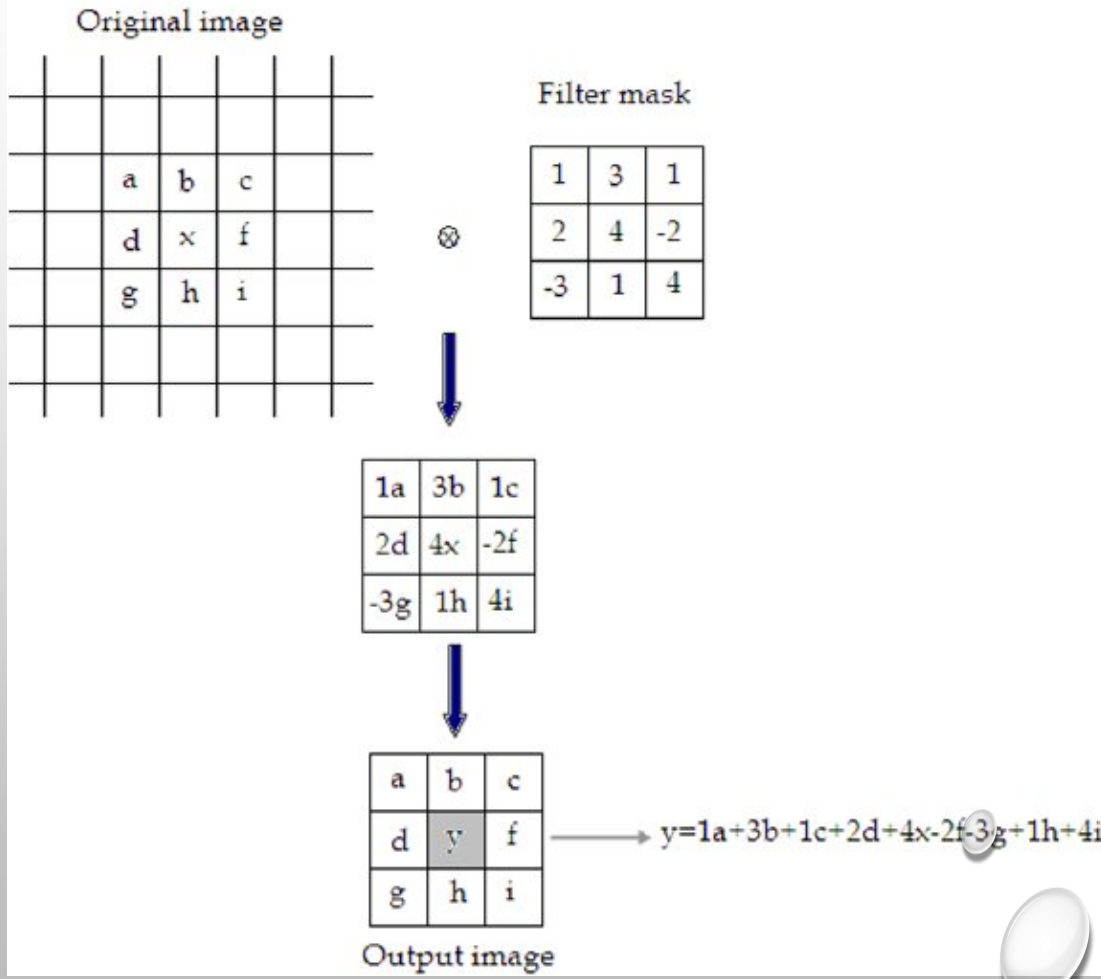
$$f * w \qquad f \otimes w$$



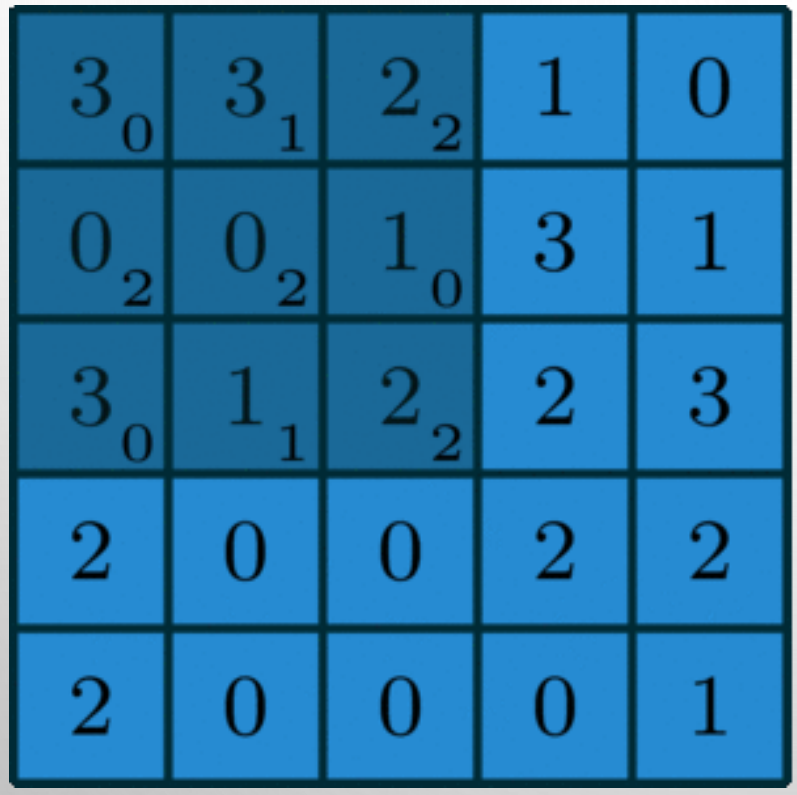
$$g(x,y) = \sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)\, f(x+s, y+t)$$

IMAGE CONVOLUTION PROCESS

- ILLUSTRATION OF FILTERING PROCESS IN SPATIAL DOMAIN. MEAN FILTER MEAN FILTER IS THE SIMPLEST LOW PASS LINEAR FILTER. IT IS IMPLEMENTED BY REPLACING EACH PIXEL VALUE WITH THE AVERAGE VALUE OF ITS NEIGHBORHOOD. MEAN FILTER CAN BE CONSIDERED AS A CONVOLUTION FILTER. THE SMOOTHING EFFECT DEPENDS ON THE KERNEL SIZE. AS THE KERNEL SIZE INCREASES, THE SMOOTHING EFFECT INCREASES TOO. USUALLY A 3 × 3 (OR LARGER) KERNEL FILTER IS USED. AN EXAMPLE OF A SINGLE 3 × 3 KERNEL

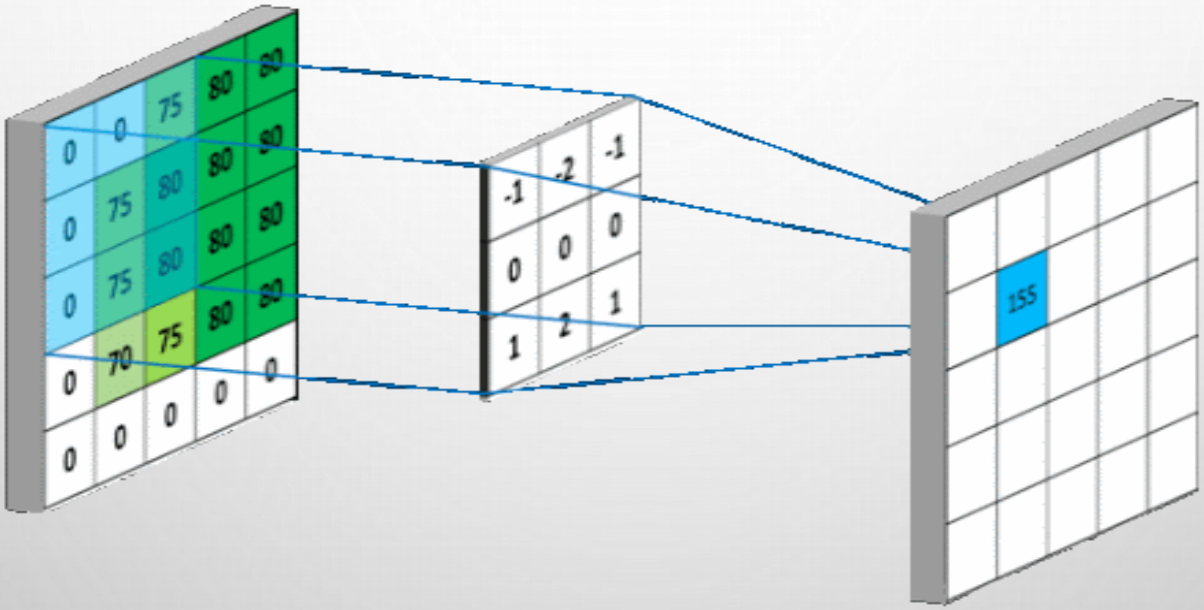


WHAT IS CONVOLUTION: BASIC OPERTION?

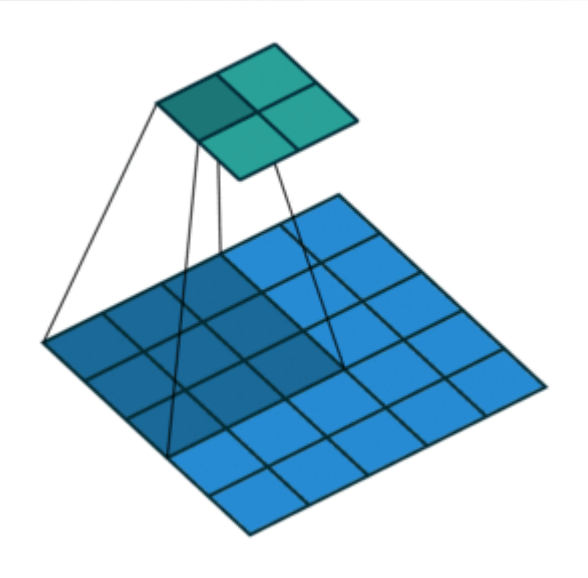


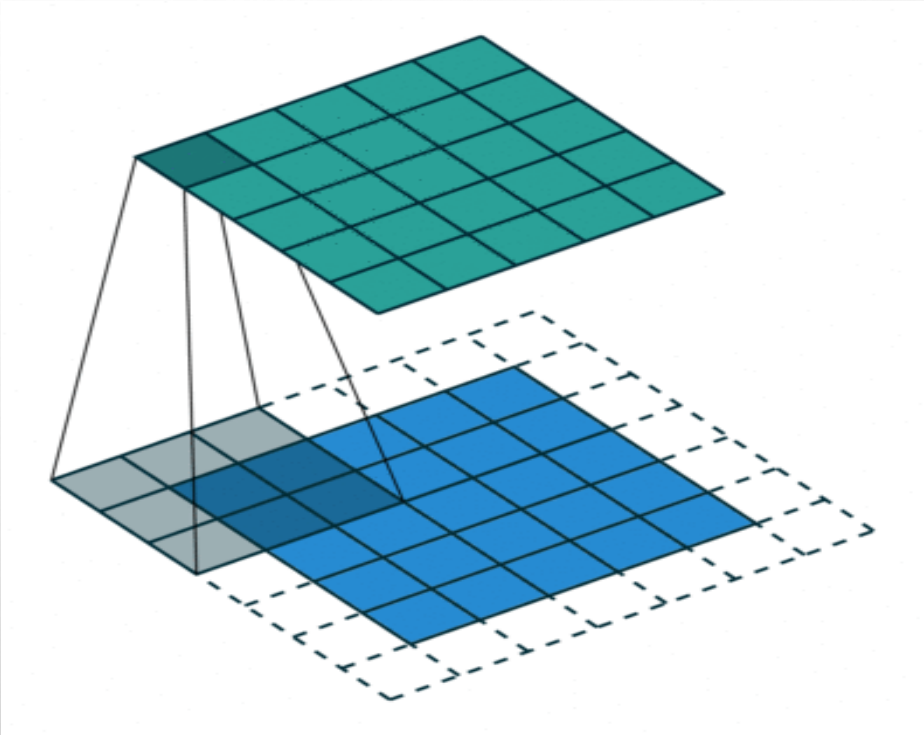


WHAT IS CONVOLUTION: BASIC OPERTION?



WHAT IS CONVOLUTION 2?
OUTPUT IMAGE SIZE:





EXAMPLE1: AVERAGING

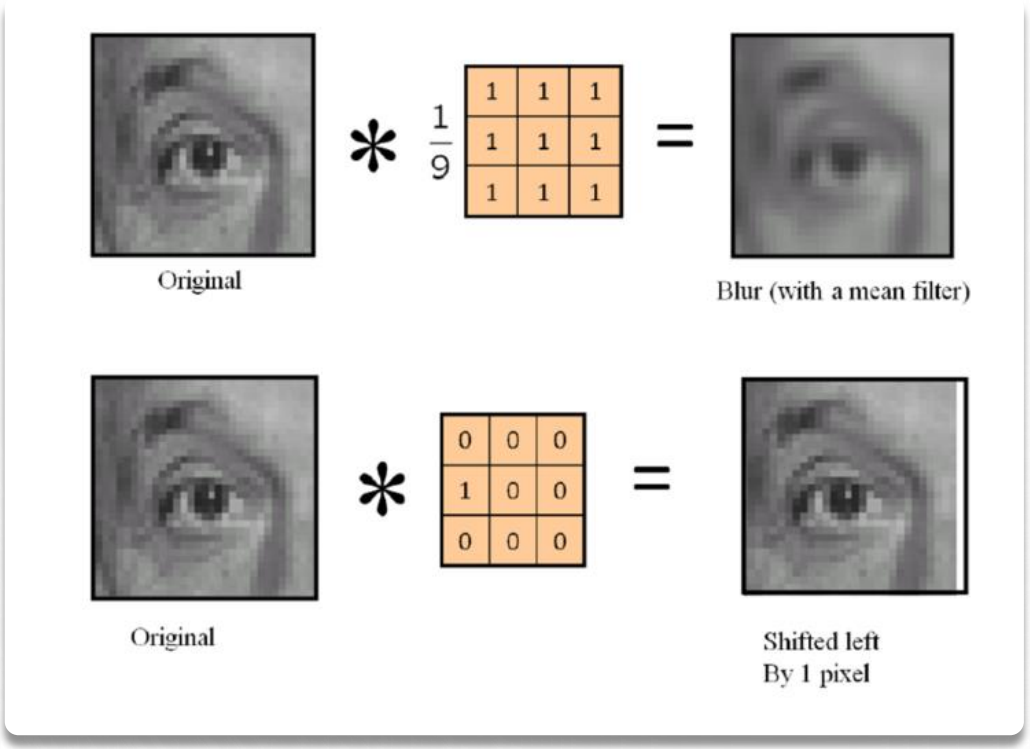
- ONE SIMPLE EXAMPLE IS SMOOTHING USING A 3X3 MASK.

f(x-1,y-1)	f(x-1,y)	f(x-1,y+1)
f(x,y-1)	f(x,y)	f(x,y+1)
f(x+1,y-1)	f(x+1,y)	f(x+1,y+1)

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

$$g(x,y) = \frac{1}{9}\left[f(x-1,y-1) + f(x-1,y) + f(x-1,y+1) + \ldots + f(x+1,y+1) \right]$$

EXAMPLES


Original
1 1 1
1 1 1
1 1 1
1/9
Blur (with a mean filter)
Original
0 0 0
1 0 0
0 0 0
Shifted left
By 1 pixel

EXAMPLE AVERAGE SMOOTHING FILTER

7	8	8	8	6	4	2	2
9	8	10	10	8	4	4	2
9	6	9	9	7	6	6	
7	7	9	9	8	8		
7	7	8	8	8			

7	7	7	6	4	3
8	8	8	7	5	4
8	9	9	8	6	
8	8	8	8		





Matlab Code

```
clear
in_image = imread('msalah 2.png');
h = fspecial('average',11);
out_image=imfilter(in_image,h);
imshow(out_image)
```

THE EFFECT OF AVERAGE FILTER SIZE



original



Filter size = 11



Filter size = 15

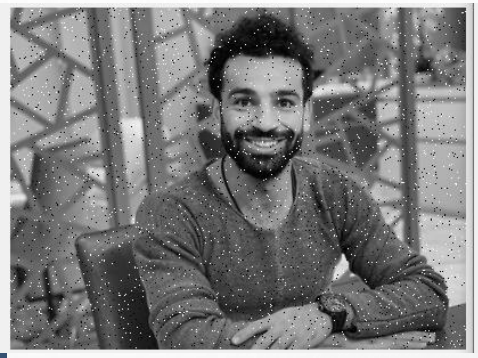


Filter size = 21

THE EFFECT OF NOISE LEVEL

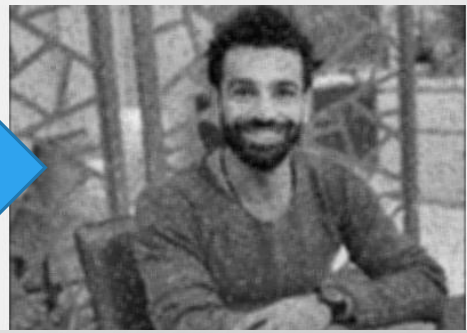


```
clear
in_image = imread('msalah
2.png');
in_image = rgb2gray(in_image);
in_image2 =
imresize(in_image,0.4);
imshow(in_image2),figure
noised_image =
imnoise(in_image2,'salt &
pepper',0.2)
imshow(noised_image),figure
h = fspecial('average',5);
out_image=imfilter(noised_image,h);
imshow(out_image)
```

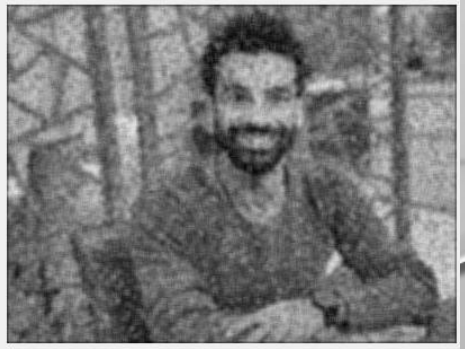










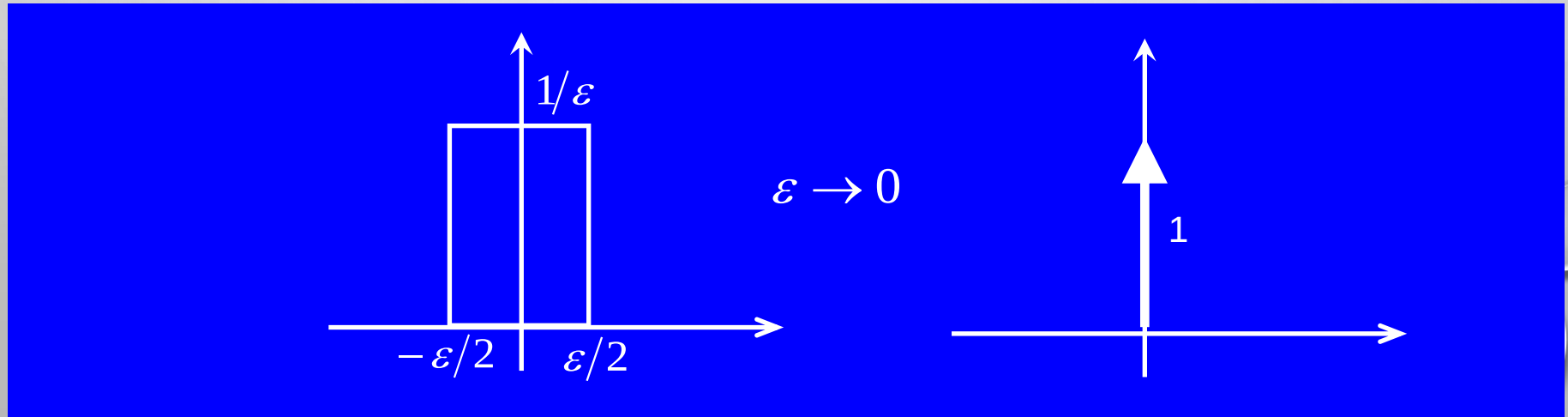


EXAMPLE2: DELTA/IMPULSE FUNCTION

- AN IDEAL IMPULSE IS DEFINED USING THE DIRAC DISTRIBUTION $\delta(x, y)$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\delta(x,y)dxdy = 1, \text{ and } \delta(x,y) = 0 \forall x, y \neq 0$$

- TO VISUALIZE IN 1D, PICTURE A RECTANGULAR PULSE FROM $x-\varepsilon/2$ TO $x+\varepsilon/2$ WITH A HEIGHT OF $1/\varepsilon$. AS $\varepsilon \to 0$, THE WIDTH TENDS TO 0 AND THE HEIGHT TENDS TO INFINITY AS THE TOTAL AREA REMAINS CONSTANT AT 1.



(CONT)

- SIFTING PROPERTY: IT PROVIDES THE VALUE OF F(X,Y) AT THE POINT (A,B)

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\delta(x-a,y-b)dxdy = f(a,b)$$

- DELTA FUNCTION AS THE MASK.

$$g(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x-s,y-t)\delta(s,t)dsdt = f(x,y)$$

  - WHEN MOVING ACROSS AN IMAGE, IT "COPIES" THE VALUE OF IMAGE INTENSITY.

GENERAL FORMULATION

- 2D CONTINUOUS SPACE:

$$g(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x-s, y-t)w(s,t)dsdt$$

- 2D DISCRETE SPACE:

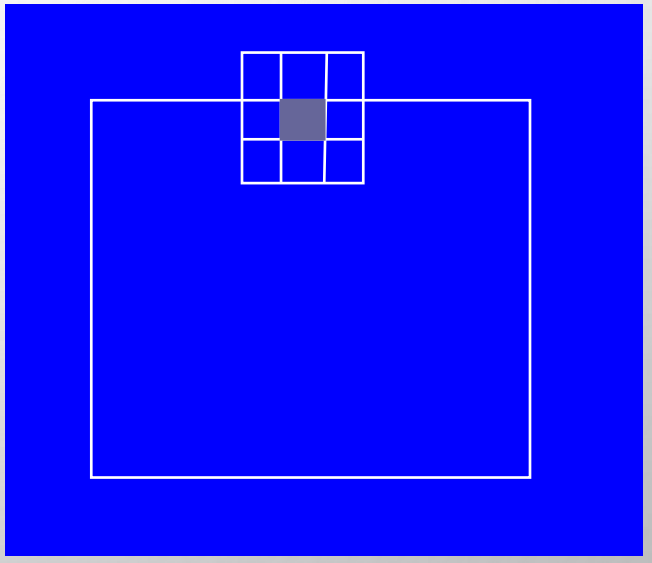
$$g[x,y] = \sum_{s=-\infty}^{\infty}\sum_{t=-\infty}^{\infty} f[x-s, y-t]w[s,t]$$

Convolution kernel/mask

- IN DIGITAL IMAGE PROCESSING, WE ONLY DEAL WITH DISCRETE SPACE WITH A MASK EFFECTIVE LOCALLY.
- CONVOLUTION IS COMMUTATIVE, ASSOCIATIVE AND DISTRIBUTIVE.

WHAT HAPPENS AT THE BORDERS?

- THE MASK FALLS OUTSIDE THE EDGE.
- SOLUTIONS?
  - IGNORE THE EDGES
    - THE RESULTANT IMAGE IS SMALLER THAN THE ORIGINAL
  - PAD WITH ZEROS
    - INTRODUCING UNWANTED ARTIFACTS



VALUES OUTSIDE THE RANGE

- LINEAR FILTERING MIGHT BRING THE INTENSITY OUTSIDE THE DISPLAY RANGE.
- SOLUTIONS?
  - CLIP VALUES

$$y = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \le x \le 255 \\ 255 & \text{if } x > 255 \end{cases}$$

  - SCALING TRANSFORMATION

$$y = 255\frac{x - g_L}{g_H - g_L}$$

New min

New max

TRANSFORM VALUES IN $[G_L, G_H]$ TO [0,255]

SEPARABLE FILTERS

- SOME FILTERS CAN BE IMPLEMENTED BY SUCCESSIVE APPLICATION OF TWO SIMPLER FILTERS.

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\ 1\\ 1\end{bmatrix}\frac{1}{3}\begin{bmatrix}1 & 1 & 1\end{bmatrix}$$

- SEPARABILITY RESULTS IN GREAT TIME SAVING.
- BY HOW MUCH?
  - FOR A MASK OF SIZE NXN, FOR EACH IMAGE PIXEL
    - ORIGINALLY, N^2 MULTIPLICATIONS AND N^2-1 ADDITIONS
    - AFTER SEPARATION, 2N MULTIPLICATIONS AND 2N-2 ADDITIONS

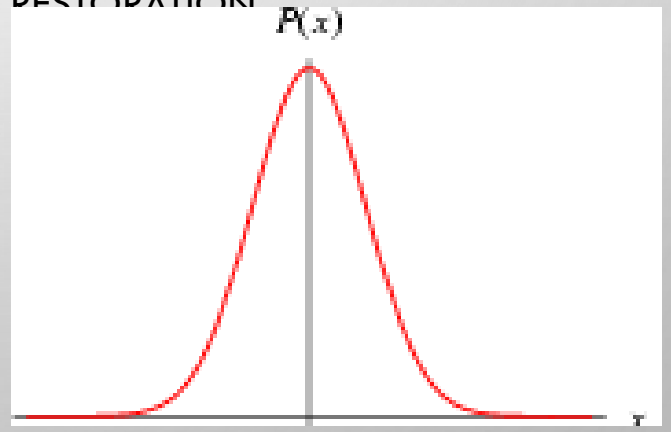
SOME TERMINOLOGY ON FREQUENCY…

- FREQUENCIES: A MEASURE OF THE AMOUNT BY WHICH GRAY VALUES CHANGE WITH DISTANCE.
- HIGH/LOW-FREQUENCY COMPONENTS: LARGE/SMALL CHANGES IN GRAY VALUES OVER SMALL DISTANCES.
- HIGH/LOW-PASS FILTER: PASSING HIGH/LOW COMPONENTS, AND REDUCING OR ELIMINATING LOW/HIGH-FREQUENCY FILTER.

LOW-PASS FILTER

- MOSTLY FOR NOISE REDUCTION/REMOVAL AND SMOOTHING
  - 3X3 AVERAGING FILTER TO BLUR EDGES
  - GAUSSIAN FILTER,
    - BASED ON GAUSSIAN PROBABILITY DISTRIBUTION FUNCTION
    - A POPULAR FILTER FOR SMOOTHING
    - MORE LATER WHEN WE DISCUSS IMAGE RESTORATION

In 1D: $P(x) = e^{-x^2/2\sigma^2}$



HIGH-PASS FILTER

- THE LAPLACIAN, $\nabla^2 f(x,y)$, IS A HIGH-PASS FILTER,

$$h_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{or} \quad h_2 = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

- SUM OF COEFFICIENTS IS ZERO
- INSIGHT: IN AREAS WHERE THE GRAY VALUES ARE SIMILAR (LOW-FREQUENCY), APPLICATION OF THE FILTER MAKES THE GRAY VALUES CLOSE TO ZERO.

LAPLACIAN OPERATOR

- LAPLACIAN IS A DERIVATIVE OPERATOR – 2ND ORDER DERIVATIVE

$$\nabla^2 f(x,y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\left.\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= f(x+1,y) + f(x-1,y) - 2f(x,y) \\ \frac{\partial^2 f}{\partial y^2} &= f(x,y+1) + f(x,y-1) - 2f(x,y) \end{aligned}\right\} \Rightarrow h_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- HIGHLIGHTING GRAY LEVEL DISCONTINUITIES.
- AN EDGE DETECTOR
- ISOTROPIC IN X- AND Y-DIRECTION.

LAPLACIAN OPERATOR (CON'D)

- WE MAY ADD THE DIAGONAL TERMS TOO => ISOTROPIC RESULTS FOR INCREMENTS OF 45°

$$h_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{or} \quad h_4 = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

- THERE ARE OTHER VARIATIONS/APPROXIMATIONS:

$$h_1 = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

Separable filter

LAPLACIAN FILTER: DEMO

Input image



Convolution Kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Feature map



Original

Laplacian-filtered

LAPLACIAN FILTER: DEMO



Original



Laplacian-filtered

LAPLACIAN: APPLICATION

- LAPLACIAN HIGHLIGHTS DISCONTINUITIES, AND TURN FEATURELESS REGIONS INTO DARK BACKGROUND.
- HOW CAN WE MAKE GOOD USE OF THE PROPERTY?
  - ONE EXAMPLE IS EDGE ENHANCEMENT

$$g(x,y)=\begin{cases} f(x,y)-\nabla^2 f(x,y), \text{ using } h_1 \\ f(x,y)+\nabla^2 f(x,y), \text{ using } h_2 \end{cases}$$

original



Laplacian enhanced

MORE FILTERS …

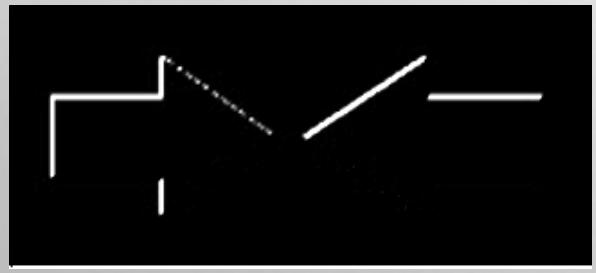
- WHAT FEATURE DOES EACH MASK HIGHLIGHT?

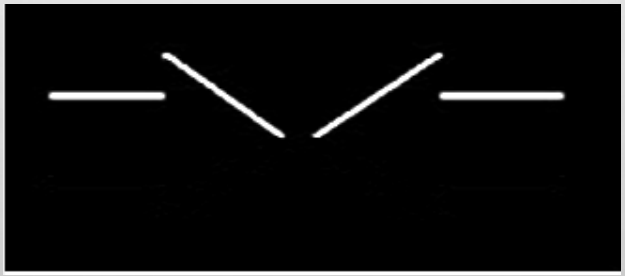


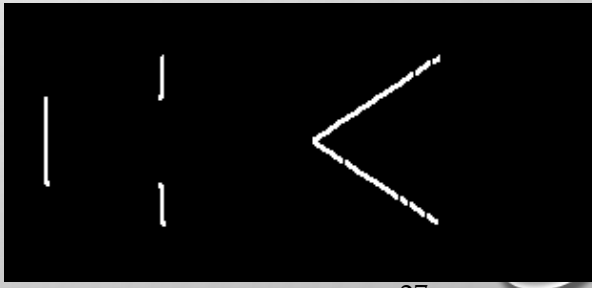
$$h_1 = \begin{bmatrix} -1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$h_2 = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

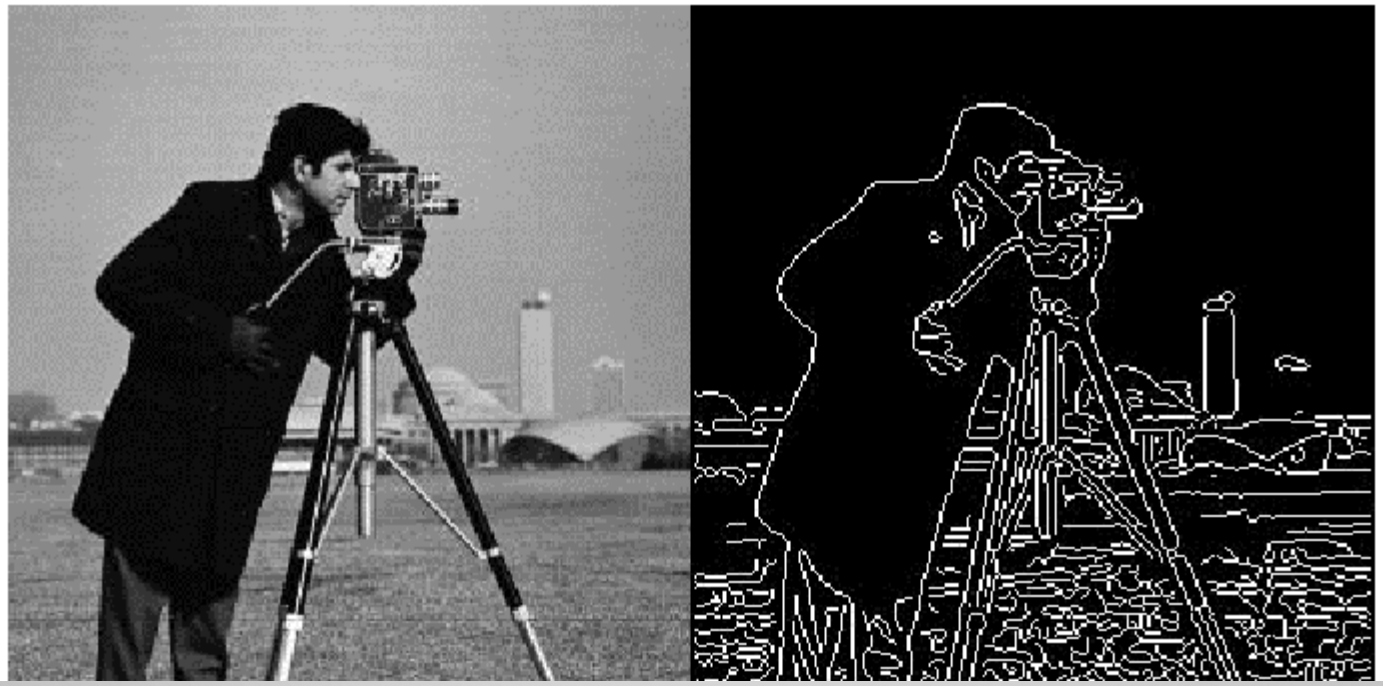
$$h_3 = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$







EXAMPLE



EDGE SHARPING

- TO MAKE EDGES SLIGHTLY SHARPER AND CRISPER.
- THIS OPERATION IS REFERRED TO AS EDGE ENHANCEMENT, EDGE CRISPENING OR UNSHARP MASKING.
- A VERY POPULAR PRACTICE IN INDUSTRY.
- SUBTRACTING A BLURRED VERSION OF AN IMAGE FROM THE IMAGE ITSELF.

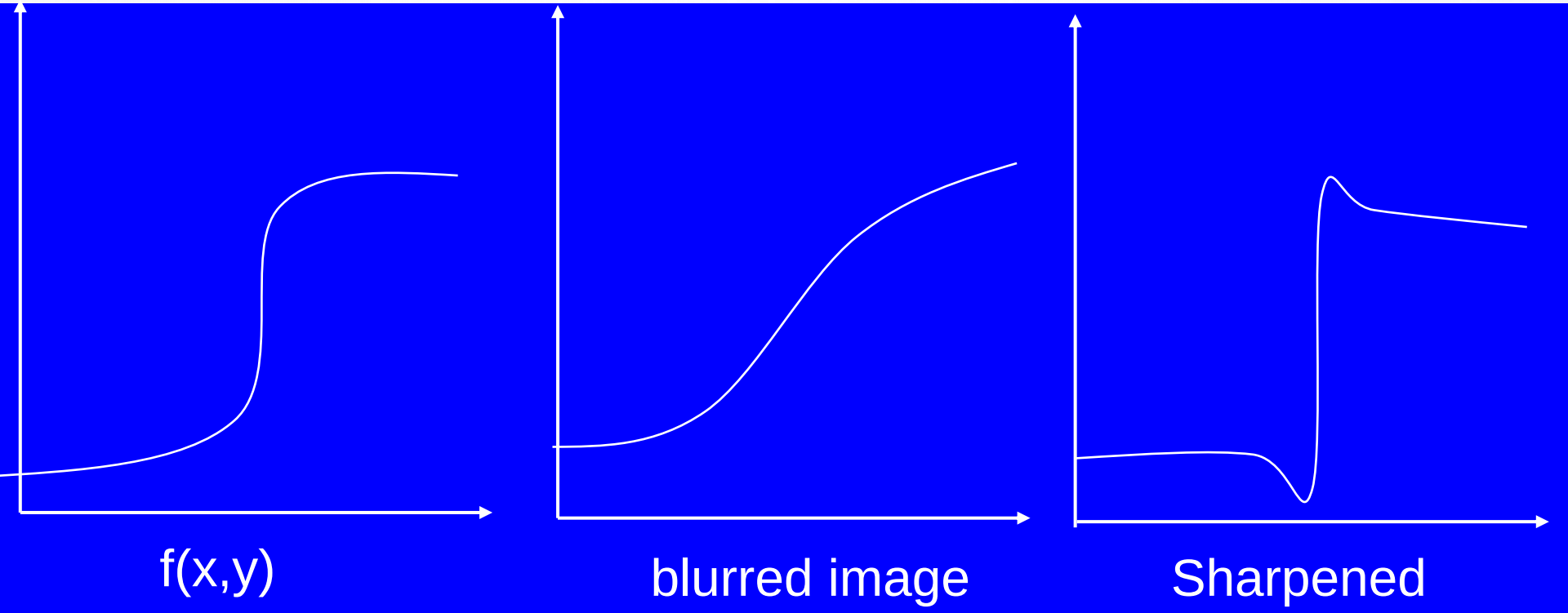
$$f_s(x,y) = f(x,y) - \overline{f}(x,y)$$

Sharpened image

Original image

Blurred image

UNSHARP MASKING: WHY IT WORKS

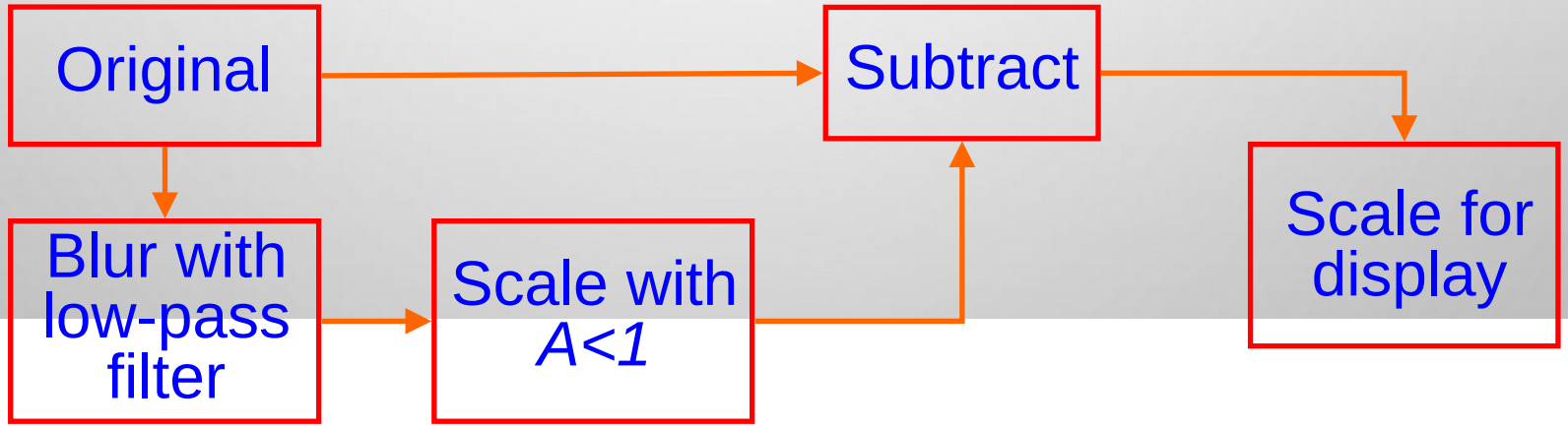


FILTER FOR UNSHARP MASKING

- COMBINING FILTERING AND SUBTRACTING IN ONE FILTER.

$$h = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{A}\begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} \approx A\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$$

- SCHEMATICALLY,



HIGH-BOOST FILTERING

- GENERALIZATION OF UNSHARP MASKING

$$f_{hb}(x,y) = Af(x,y) - \overline{f}(x,y)$$

- HERE A IS CALLED BOOST COEFFICIENT, AND

$$A \geq 1$$

- WE REWRITE THE EQUATION AS

$$\begin{aligned} f_{hb}(x,y) &= (A-1)f(x,y) + f(x,y) - \overline{f}(x,y) \\ &= (A-1)f(x,y) + f_s(x,y) \end{aligned}$$

f_s $\qquad$ $f \pm \nabla^2 f$

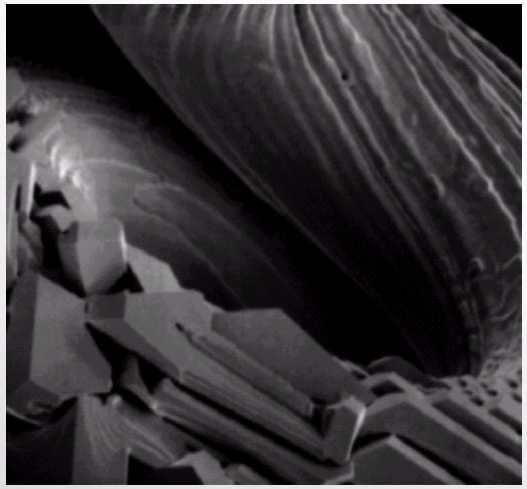
$$h = \begin{bmatrix} 0 & -1 & 0 \\ -1 & A+4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

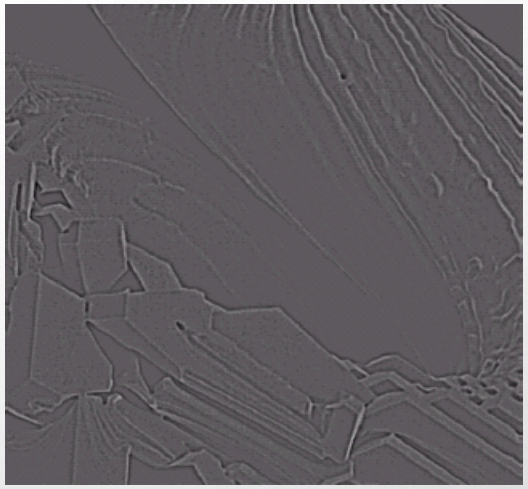
$$h = \begin{bmatrix} 0 & -1 & 0 \\ -1 & A+4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

HIGH-BOOST FILTERING: DEMO

- BY VARYING A, BETTER OVERALL BRIGHTNESS CAN BE IMPROVED.

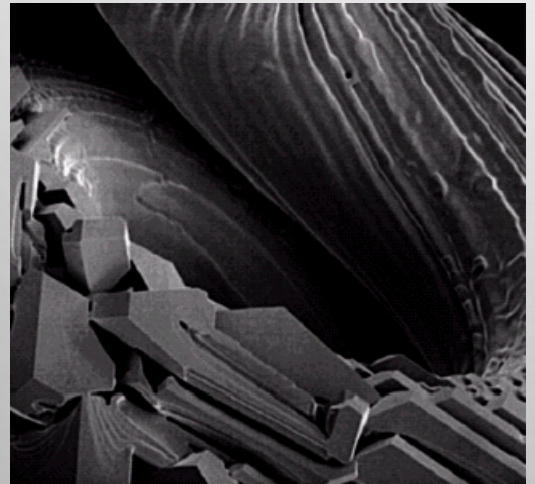
original

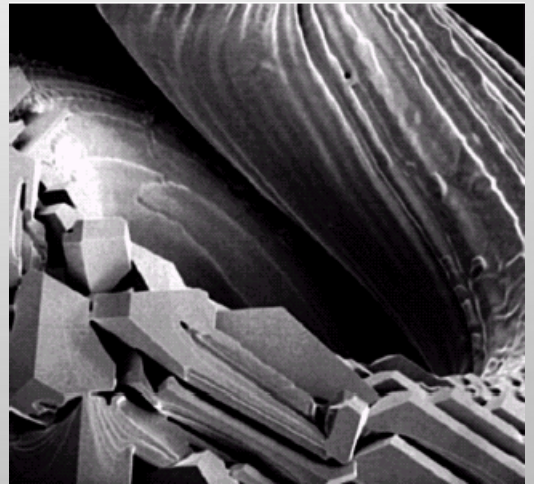




A=0

A=1



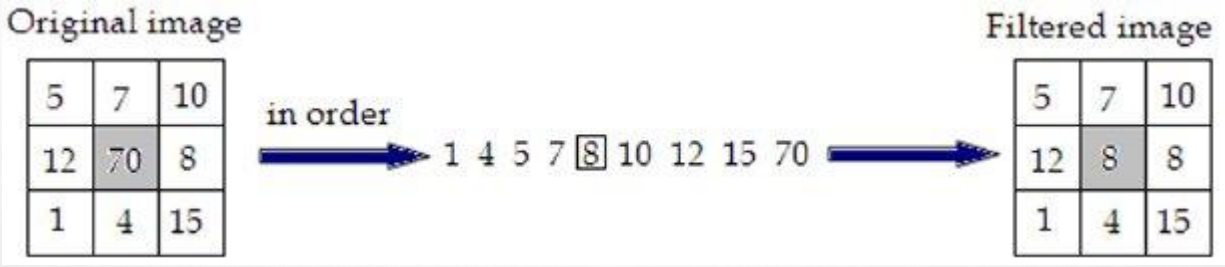


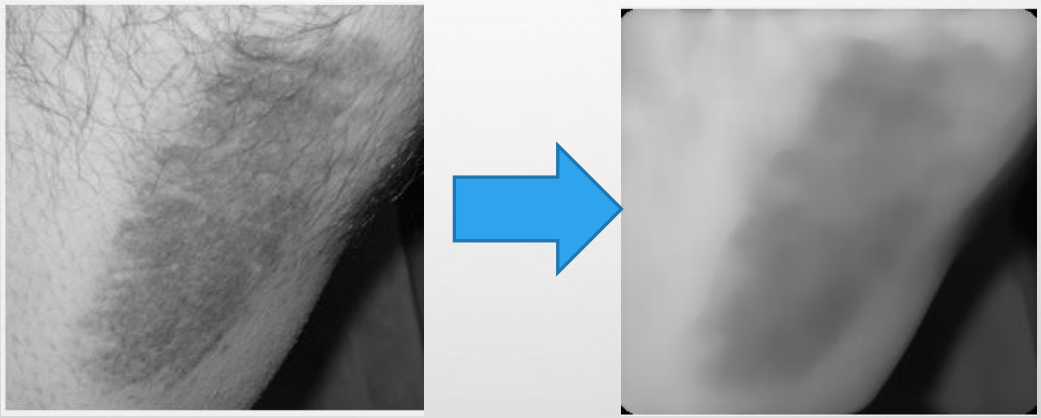
A=1.7, much brighter

NONLINEAR FILTERS

- WILL DISCUSS SOME OF THEM IN MORE DETAIL LATER FOR THE PURPOSE OF IMAGE RESTORATION.
- MAXIMUM FILTER: THE OUTPUT THE MAXIMUM VALUE UNDER THE MASK
- MINIMUM FILTER: THE OUTPUT THE MINIMUM VALUE UNDER THE MASK
- RANK-ORDER FILTER:
  - ELEMENTS UNDER THE MASK ARE ORDERED, AND A PARTICULAR VALUE IS RETURNED.
  - BOTH MAXIMUM AND MINIMUM FILTER ARE INSTANCES OF RANK-ORDER
  - ANOTHER POPULAR INSTANCE IS MEDIAN FILTER

FILTERING APPROACH OF MEDIAN FILTER





```
clear
in_image = imread('h_skin.jpg');
in_image = rgb2gray(in_image);
in_image = imresize(in_image,0.6);
imshow(in_image),figure
out_image=medfilt2(in_image,[21,21]);
imshow(out_image)
```

MORE NONLINEAR FILTERS

- ALPHA-TRIMMED MEAN FILTER:
  - ORDER THE VALUES UNDER THE MASK
  - TRIM OFF ELEMENTS AT EITHER END OF THE ORDERED LIST
  - TAKE THE MEAN OF THE REMAINDER
  - E.G. ASSUMING A 3X3 MASK AND THE ORDERED LIST

$$x_1 \leq x_2 \leq x_3 \leq \cdots \leq x_9$$

  - TRIMMING OF TWO ELEMENTS AT EITHER END, THE RESULT OF THE FILTER IS

$$\left(x_3 + x_4 + x_5 + x_6 + x_7\right)/5$$

SUMMARY

- WE INTRODUCED THE CONCEPT OF CONVOLUTION
- WE BRIEFLY DISCUSSED SPATIAL FILTERS OF
  - LOW-PASS FILTER FOR SMOOTHING
  - HIGH-PASS FILTER FOR EDGE SHARPENING
  - NONLINEAR
- WE'LL COME BACK TO MOST OF THE FILTERS UNDER THE APPROPRIATE TOPICS.